

MCA SEMESTER-I
School of Computer and Systems Sciences,
Jawaharlal Nehru University,
New Delhi -110067
Computer Fundamental (CS-402) Mid-I
Date: 08.09.2024

Total Marks: 20

Time: 2hr

All Compulsory

1. Convert the Hexadecimal 2AC5 to decimal, octal and Binary bases. 2
2. Simplify the Boolean function to a minimum number of literals: $x'y' + xy + x'y$ 2
3. Find the complement of : $(x+y'+z)(x'+z')(x+y)$ 2
4. Implement half subtractor. 2
5. Simplify the function: $F = \Sigma(1, 4, 5, 6, 12, 14, 15)$. Implement the function without simplification and after simplifying to check the number of gates in each case. 4
6. Implement BCD to excess three code converter. 4
7. Implement Full adder using decoder. 4

MCA SEMESTER-I
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End sem-I
Date: 20.12.2024@10am

Total Marks: 50

Time: 3hr

All Compulsory

1. Write note on
 - a. Parity checker and generator
 - b. The need of minimization in a logic circuit
 - c. Combinational and sequential circuits
 - d. Any two ID digital logic families
 - e. Edge triggering and Level triggering
2. Represent decimal number 8620 in
 - a. BCD
 - b. Excess-3 code
 - c. 2421
 - d. Binary
 - e. Hexadecimal
3. What will be the output signal f and g for logic circuit shown in fig 1 with following a, b and c input signals shown in Fig 2?

3*5=15

5

5

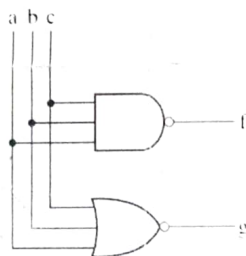


Fig 1 : Logic Circuit

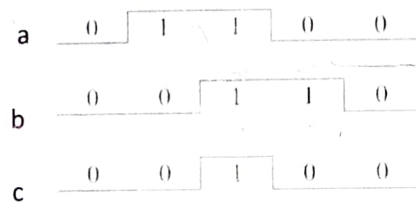


Fig 2 : Input Signal

4. Simplify the Boolean Function F together with don't care conditions d in (i) sum of product and (ii) Product of sum

$$F(w, x, y, z) = \Sigma(0, 1, 2, 3, 7, 8, 10)$$

$$d(w, x, y, z) = \Sigma(5, 6, 11, 15)$$
5. Implement a 3 bit magnitude comparator.
6. Implement the following functions with PLA

$$F_1(A, B, C) = \Sigma(3, 5, 6, 7)$$

$$F_2(A, B, C) = \Sigma(0, 2, 4, 7)$$
7. a) What is a flip flop? Explain the T-flip flop alongwith state table, characteristic table with characteristic equation.
 b) Implement 3 bit step-up counter using T flip. Make state diagram, excitation table and logic circuit.

4+6